

Budget Justification

Computational Resources: \$5,000 USD.

- **Computer equipment (\$3,000 USD):**

Funds are requested for the purchase of dedicated computer equipment to support the Principal Investigator (PI) and a graduate student working on Aims A1 and A2. Additional components (e.g., memory upgrades) may be included. These resources are essential for developing the proposed MOD-VAR testing procedures and the associated R software package. This package is critical to ensuring broad access and adoption of the proposed methods within the quantitative psychology research community.

- **High-Performance Computing Center (HPCC) subscription (\$2,000 USD):**

Funds are requested to support a 24-month subscription (July 2026 – June 2028) to the University of California, Riverside High-Performance Computing Center (HPCC), in support of Aim A2. Access to HPCC resources is necessary to conduct large-scale simulation studies designed to evaluate the performance of the proposed methods under realistic data conditions. These studies require the execution of thousands of simulations across a wide range of parameter settings and competing methods. Therefore, there is a need for parallel processing and optimization capabilities beyond standard personal computing systems.

Travel: \$4,000 USD.

- **Conference travel (\$2,000 USD):**

Funds are requested to support the PI's participation in the Joint Computational and Methodological Statistics Conference (CMStatistics), to be held at the University of Applied Sciences in Berlin, Germany, in December 2026. The PI will present the invited talk, *"Estimating Moderator Effects on Multiple-Subject Granger-Causal Networks on Intensive Longitudinal Data,"* and will organize a session titled *"Emerging Methods in High-Dimensional Multi-Subject Longitudinal Data."* Funds will support registration, airfare, and lodging costs associated with the conference. This activity will support dissemination of project findings and is in alignment with Aim A3.

- **University visits (\$2,000 USD):**

Funds are requested to partially support academic visits to collaborating institutions to advance the proposed research and establish new partnerships. In particular, the PI will coordinate a research visit with Prof. Marie Düker at the Technical University of Munich (TUM), where the proposed methods will be applied to intensive multi-subject neuroimaging data. This visit is expected to result in collaborative analyses and joint research outputs, in alignment with Aim A3.

Total Direct Costs – \$9,000 USD.